

Least-squares fitting of a constant attenuation target

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Abstract

1 Problem statement

To use L SLS mechanisms to fit nearly constant Q over a frequency band $[\omega_{\min}, \omega_{\max}]$, we minimize the integrated squared misfit (Yang et al., 2016)

$$\chi_0(Y) = \int_{\omega_{\min}}^{\omega_{\max}} \left(\tilde{Q}^{-1}(\omega; Y) - Q^{-1} \right)^2 d\omega. \quad (1)$$

where

$$\tilde{Q}^{-1}(\omega; Y) = \frac{\sum_{\ell=1}^L Y_{\ell} a_{\ell}(\omega)}{1 - \sum_{\ell=1}^L Y_{\ell} b_{\ell}(\omega)}, \quad a_{\ell}(\omega) = \frac{\omega_{\ell} \omega}{\omega_{\ell}^2 + \omega^2}, \quad b_{\ell}(\omega) = \frac{\omega_{\ell}^2}{\omega_{\ell}^2 + \omega^2}. \quad (2)$$

The unknowns are the dimensionless anelastic coefficients $Y = (Y_1, \dots, Y_L)^{\top}$, subject to the box constraint

$$0 \leq Y_{\ell} \leq 1, \quad \ell = 1, \dots, L. \quad (3)$$

2 Quadrature discretization

We evaluate (1) on M logarithmically spaced frequency nodes. With $s = \ln \omega$, the change of variables $d\omega = \omega ds$ gives the nodes

$$s_m = \ln \omega_{\min} + \frac{m-1}{M-1} (\ln \omega_{\max} - \ln \omega_{\min}), \quad m = 1, \dots, M, \quad (4)$$

so that $\omega_m = e^{s_m}$. Applying the trapezoidal rule in s , with $\Delta s = (\ln \omega_{\max} - \ln \omega_{\min}) / (M-1)$, yields the weights

$$w_m = \Delta s \omega_m \begin{cases} \frac{1}{2}, & m = 1 \text{ or } m = M, \\ 1, & \text{otherwise.} \end{cases} \quad (5)$$

The factor ω_m accounts for the change of variables, so the objective measures misfit with respect to $d\omega$, rather than $d \ln \omega$. The discrete objective is

$$\chi_0(Y) \approx \sum_{m=1}^M w_m \left(\tilde{Q}^{-1}(\omega_m; Y) - Q^{-1} \right)^2, \quad \text{subject to (3)}. \quad (6)$$

To compute the analytic Jacobian, define at each node m

$$N_m = \sum_{\ell=1}^L Y_{\ell} a_{\ell}(\omega_m), \quad D_m = 1 - \sum_{\ell=1}^L Y_{\ell} b_{\ell}(\omega_m), \quad \tilde{Q}_m^{-1} = \frac{N_m}{D_m}. \quad (7)$$

so that

$$\frac{\partial \tilde{Q}_m^{-1}}{\partial Y_{\ell}} = \frac{a_{\ell}(\omega_m) + \tilde{Q}_m^{-1} b_{\ell}(\omega_m)}{D_m}. \quad (8)$$

leading to the Jacobian $K(Y) \in \mathbb{R}^{M \times L}$ with entries

$$K_{m\ell}(Y) = \sqrt{w_m} \frac{a_\ell(\omega_m) + \tilde{Q}_m^{-1} b_\ell(\omega_m)}{D_m}, \quad m = 1, \dots, M, \ell = 1, \dots, L. \quad (9)$$

3 Projected Levenberg–Marquardt iteration

The code uses a projected Levenberg–Marquardt iteration. At the current iterate $Y^{(k)}$, define

$$A_k = K(Y^{(k)})^\top K(Y^{(k)}), \quad b_k = K(Y^{(k)})^\top r(Y^{(k)}), \quad S_k = \text{diag}(1 + (A_k)_{11}, \dots, 1 + (A_k)_{LL}). \quad (10)$$

Starting with $\lambda = 10^{-3}$, each iteration proceeds as follows:

1. Compute $r(Y^{(k)})$, $K(Y^{(k)})$, A_k , b_k , and $\chi_h(Y^{(k)})$. Solve the damped normal equations

$$(A_k + \lambda S_k) \delta = -b_k. \quad (11)$$

2. Project the trial coefficients componentwise onto $[0, 1]^L$:

$$\hat{Y}_\ell = \min(1, \max(0, Y_\ell^{(k)} + \delta_\ell)). \quad (12)$$

3. If $\chi_h(\hat{Y}) < \chi_h(Y^{(k)})$, accept $Y^{(k+1)} = \hat{Y}$ and set $\lambda \leftarrow \max(0.3\lambda, 10^{-12})$. Otherwise, set $\lambda \leftarrow \min(3\lambda, 10^{12})$ and recompute the trial step using the same A_k , b_k , and S_k . At most 30 trial steps are attempted per iteration.
4. After the trial loop, stop if no step was accepted or if $\|b_k\|_2 < 10^{-12}$. Each run is also limited to 200 iterations.

References

Yang, P., Brossier, R., Métivier, L., and Virieux, J. (2016). A review on the systematic formulation of 3D multiparameter full waveform inversion in viscoelastic medium. *Geophysical Journal International*, 207(1):129–149.